数学笔记

BeBop

October 26, 2025

Contents

Chapter 1

高等代数

Problem 1.0.1. 假设 n 维欧氏空间 V 中有 m 个向量两两成钝角, 证明: $m \leqslant n+1$.

Proof. 当 $\dim V = 1$, 命题显然成立.

假设当维数为 $n-1$ 时命题成立, 现证明命题在 n 维时仍成立. 设 $v_1, \ldots, v_m$ 为 V 中两两成钝角的向量, 令 W 为 Lv_m 的正交补空间, 并记 $P: V \to W$ 为正交投影映射, 令 $w_1 = Pv_1, \ldots, w_{m-1} = Pv_{m-1}$, 则当 $i \neq j$ 时,

$$\begin{aligned}
\langle w_i, w_j \rangle = \langle Pv_i, Pv_j \rangle &= \left\langle v_i - \frac{\langle v_i, v_m \rangle}{\langle v_m, v_m \rangle} v_m, v_j - \frac{\langle v_j, v_m \rangle}{\langle v_m, v_m \rangle} v_m \right\rangle \\
&= \langle v_i, v_j \rangle - \frac{\langle v_i, v_m \rangle \cdot \langle v_j, v_m \rangle}{\langle v_m, v_m \rangle} \\
&< 0
\end{aligned}$$

因此 $w_1, \ldots, w_{m-1}$ 是 $n-1$ 维空间 W 中两两成钝角的向量, 由归纳假设知 $m-1 \leqslant n$, 即 $m \leqslant n+1$. □

Remark. 我感觉这和 n 维欧氏空间的拓扑性质有关系,

Problem 1.0.2. Let V be an n-dimensional vector space over the field F, then $\alpha \in \bigwedge^2(V^*)$ is decomposable, i.e., there exist $\omega, \eta \in V^*$ such that $\alpha = \omega \wedge \eta$, if and only if $\alpha \wedge \alpha = 0$.

Proof. Necessity is obvious. For sufficiency, we need a lemma:

Lemma 1.0.3. *For $\alpha \in \bigwedge^r(V^*)$, $\omega \in V^*$, there exists $\eta \in \bigwedge^{r-1}(V^*)$ such that $\omega \wedge \eta = \alpha$ if and only if $\omega \wedge \alpha = 0$.*

If $\alpha \wedge \alpha = 0$, then

$$(\iota_X \alpha) \wedge \alpha = \frac{1}{2}\iota_X(\alpha \wedge \alpha) = 0, \quad \forall X \in V.$$

Since $\alpha \neq 0$, there exists $X \in V$ such that $\omega = \iota_X \alpha \neq 0$. Then by the lemma, there exists $\eta \in \bigwedge^1(V^*)$ such that

$$\omega \wedge \eta = \alpha.$$

□

Proof of lemma. Necessity is obvious.

For sufficiency, let $\omega_1, \ldots, \omega_n$ be a basis of V^* with $\omega_1 = \omega$. Then any $\alpha \in \bigwedge^r(V^*)$ can be written as

$$\alpha = \sum_{1 \leqslant i_1 < \cdots < i_r \leqslant n} a_{i_1 \cdots i_r} \omega_{i_1} \wedge \cdots \wedge \omega_{i_r}.$$

Then

$$\omega \wedge \alpha = \sum_{2 \leqslant i_1 < \cdots < i_r \leqslant n} a_{i_1 \cdots i_r} \omega_1 \wedge \omega_{i_1} \wedge \cdots \wedge \omega_{i_r} = 0,$$

which implies that $a_{i_1 \cdots i_r} = 0$ for any $1 < i_1 < \cdots < i_r \leqslant n$. Thus

$$\alpha = \sum_{2 \leqslant i_2 < \cdots < i_r \leqslant n} a_{1 i_2 \cdots i_r} \omega_1 \wedge \omega_{i_2} \wedge \cdots \wedge \omega_{i_r} = \omega \wedge \eta,$$

where

$$\eta = \sum_{2 \leqslant i_2 < \cdots < i_r \leqslant n} a_{1 i_2 \cdots i_r} \omega_{i_2} \wedge \cdots \wedge \omega_{i_r}.$$

□